

CS301 Computer Architecture

Lab 1

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1 Comparison of Instruction count among different ISAs

Q1. C program: Square of an integer

ISA and Compiler	No. of Instructions
RISC-V 32-bits gcc 14.2.0	24
RISC-V 64-bits gcc (trunk)	25
RISC-V rv32gc gcc (trunk)	N/A
RISC-V rv32gc clang(trunk)	21
RISC-V rv64gc clang(trunk)	21
x86-64 clang 12.0.0	18
x86-64 gcc 14.2.0	19
ARM64 gcc 14.2.0	19
ARM32 gcc 14.2.0	N/A
Armv8-a-clang 19.1.0	21
MIPS64 gcc 5.4	36

Q2. C program: Iterative factorial

ISA and Compiler	No. of Instructions
RISC-V 32-bits gcc 14.2.0	33
RISC-V 64-bits gcc (trunk)	35
RISC-V rv32gc gcc (trunk)	N/A
RISC-V rv32gc clang(trunk)	37
RISC-V rv64gc clang(trunk)	37
x86-64 clang 12.0.0	28
x86-64 gcc 14.2.0	27
ARM64 gcc 14.2.0	29
ARM32 gcc 14.2.0	N/A
Armv8-a-clang 19.1.0	34
MIPS64 gcc 5.4	48

Q3. C program: Sum of array elements

ISA and Compiler	No. of Instructions
RISC-V 32-bits gcc 14.2.0	70
RISC-V 64-bits gcc (trunk)	73
RISC-V rv32gc gcc (trunk)	N/A
RISC-V rv32gc clang(trunk)	75
RISC-V rv64gc clang(trunk)	75
x86-64 clang 12.0.0	48
x86-64 gcc 14.2.0	49
ARM64 gcc 14.2.0	54
ARM32 gcc 14.2.0	N/A
Armv8-a-clang 19.1.0	57
MIPS64 gcc 5.4	82

Q4. C program: Fibonacci sequence

ISA and Compiler	No. of Instructions
RISC-V 32-bits gcc 14.2.0	30
RISC-V 64-bits gcc (trunk)	32
RISC-V rv32gc gcc (trunk)	N/A
RISC-V rv32gc clang(trunk)	34
RISC-V rv64gc clang(trunk)	35
x86-64 clang 12.0.0	21
x86-64 gcc 14.2.0	21
ARM64 gcc 14.2.0	25
ARM32 gcc 14.2.0	N/A
Armv8-a-clang 19.1.0	29
MIPS64 gcc 5.4	35

Q5. C program: Array manipulation (square + 10)

ISA and Compiler	No. of Instructions
RISC-V 32-bits gcc 14.2.0	71
RISC-V 64-bits gcc (trunk)	79
RISC-V rv32gc gcc (trunk)	N/A
RISC-V rv32gc clang(trunk)	74
RISC-V rv64gc clang(trunk)	74
x86-64 clang 12.0.0	51
x86-64 gcc 14.2.0	56
ARM64 gcc 14.2.0	57
ARM32 gcc 14.2.0	N/A
Armv8-a-clang 19.1.0	60
MIPS64 gcc 5.4	97

2 Inference

Qa) For the same high-level program, the instruction sequence for different compilers and different machine architecture (represented by its ISA) - What are your observations?

Answer: For the same high-level program, the instruction sequence for different compilers and machine architecture differs mainly on the count of instructions required to achieve a task and how the data is organized. Different compilers may have different approaches to achieve the same task and hence may apply different optimization techniques which lead to different count of instructions. The organization of data is affected by the amount of available registers in a ISAs. In ISAs like RISC-V, ARM, MIPS, the Load-Store approach is followed i.e. operations on data can only be done after the data is loaded into registers while ISAs like x86 follow the Register-Memory approach and hence can operate on memory directly without first loading the data into registers.

Qb) Even for the same ISA, but for different compilers, can you comment about the generated the instruction sequences

Answer: Different compilers may compile a program in the same ISA in vastly contrasting ways. The divergence in output arises due to each compiler applying distinctive strategies to achieve a task and optimization techniques which may be motivated by the parameters that the compiler is seeking to optimize. Depending on the complexity of the program, the results may be worlds apart even if they achieve the same task.

Qc) Provide your observations about the instruction sequences generated for 32-bit and 64-bit machines of the same ISA and of the same compiler.

Answer: The instruction sequences generated for 32-bit and 64-bit machines of the same ISA and of the same compiler have a similar count of instructions. The differences between the two arise due to the data width and addressing which affects the instructions concerning how the data is loaded or stored. Hence, the instruction sequence generated for 32-bit and 64-bit machines may have similar logic while differing how data is accessed and stored.